Representational Commitment: When Hidden States Predict Behavioral Consistency in LLM Agents

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Abstract

LLM-based agents exhibit behavioral inconsistency: the same agent given the same task produces different reasoning trajectories across runs. We investigate whether this inconsistency has a signature in internal representations. Using Llama-3.1-70B on a ReAct question-answering task (100 questions, 10 runs each, 1,000 total trajectories), we find that *activation similarity*—pairwise cosine similarity of hidden states across runs at a given step—predicts behavioral consistency ($r = -0.35$, 95% CI $[-0.48, -0.17]$, permutation $p < 0.001$). Higher similarity indicates the model has not updated its representations in response to observations, remaining in an undifferentiated state from which future behavior diverges. We call this **representational commitment**: the degree to which a model integrates observations into its internal state. The signal emerges at step 4 of the agent trajectory, spans layers 32–80, and *strengthens* after controlling for task difficulty (partial $r = -0.45$, 95% CI $[-0.59, -0.29]$, $p < 10^{-5}$). Questions in the lowest activation-similarity quartile show $4.1\times$ higher behavioral variance than those in the highest quartile (Cohen's $d = 1.01$, $p = 0.002$). It is not explained by question length, context size, or other simple baselines. Notably, the signal predicts *between-run consistency* but not *per-run correctness* (probing AUC ≈ 0.5), establishing it as a fundamentally relational property of agent representations. We validate cross-model on Qwen 2.5 72B ($r = -0.65$, 95% CI $[-0.74, -0.55]$) and Phi-3-Medium-14B ($r = -0.36$, 95% CI $[-0.52, -0.17]$), a structurally different model (14B parameters, 40 layers, d=5120), confirming the finding generalizes across architectures and scales. Cross-benchmark validation on StrategyQA ($r = -0.83$, 95% CI $[-0.90, -0.76]$, n=50) confirms generalization to a different task type, with the commitment juncture shifting to step 3 consistent with shorter reasoning chains. A runtime monitor using step-4 hidden states achieves AUROC up to 0.97 for predicting consistency. A three-condition prompting intervention (control, token-matched filler, commitment) confirms that the commitment *framing*—not additional token count—reduces behavioral CV by 28% relative to the filler control ($d = 0.33$, $p = .001$) without affecting accuracy. To our knowledge, this is the first work connecting agent behavioral reliability to hidden-state geometry and demonstrating that inducing representational commitment improves agent consistency.

1 Introduction

LLM-based agents—systems combining language models with tool use and multi-step reasoning (Yao et al., 2022; Wei et al., 2022)—face a critical reliability concern: *behavioral inconsistency*. When given the same task multiple times with non-zero sampling temperature, agents produce different action sequences, explore different reasoning paths, and often reach different conclusions. This is not merely a theoretical concern: production agent systems are typically evaluated on single runs, yet the same system can produce correct answers on one run and incorrect answers on another for the same input. Inconsistency complicates deployment, debugging, and benchmarking.

Recent work shows that agent divergence concentrates at early decision points (Anonymous, 2026a) and that consistency amplifies outcomes rather than guaranteeing correctness (Anonymous, 2026b). But these studies measure what agents *do* without examining what happens *inside* the model. Does behavioral consistency have a measurable signature in internal representations?

We answer affirmatively. Extracting hidden states from Llama-3.1-70B (Grattafiori et al., 2024) during ReAct (Yao et al., 2022) question-answering on HotpotQA (Yang et al., 2018), we find that cross-run *activation similarity*—how similar hidden states are across multiple runs of the same question—predicts downstream behavioral consistency. We introduce the concept of **representational commitment**: when a model integrates observations into differentiated internal states, each run follows a focused strategy and behavior is consistent; when internal states remain undifferentiated despite receiving different observations, the model has not committed, and behavior diverges.

Our contributions:

1. Activation similarity at step 4 predicts consistency ($r = -0.35$, 95% CI $[-0.48, -0.17]$, permutation $p < 0.001$, n=100), *strengthening* after difficulty controls (partial $r = -0.45$, 95% CI $[-0.59, -0.29]$, $p < 10^{-5}$). Quartile analysis reveals a 4.1× effect: questions with lowest similarity show mean CV = 0.25 vs. 0.06 for highest ($d = 1.01$).
2. The signal has temporal and spatial specificity: absent at steps 1–2, peaking at step 4, spanning layers 32–80.
3. The signal predicts *between-run consistency* but not *per-run correctness* (probing AUC ≈ 0.5), distinguishing committed from uncommitted states but not correct from incorrect ones.
4. Cross-model validation on Qwen 2.5 72B ($r = -0.65$) and Phi-3-Medium-14B ($r = -0.36$), spanning 14B–72B parameters and three architectures, and cross-benchmark validation on StrategyQA ($r = -0.83$, n=50) confirm generalization across model families, scales, and task types.
5. A runtime monitor using step-4 hidden states achieves AUROC up to 0.97 (Llama) and 0.93 (Qwen) for predicting behavioral consistency via LOOCV, substantially outperforming surface-level baselines.
6. A three-condition prompting intervention (control, token-matched filler, commitment) isolates the commitment framing effect: the commitment prompt reduces CV by 28% relative to the filler control ($d = 0.33$, $p = .001$) without affecting accuracy, ruling out token-count confounds.

2 Related Work

Behavioral consistency in agents. Anonymous (2026a) introduced multi-run behavioral consistency as an agent reliability metric on HotpotQA, finding a 32–55 percentage point accuracy gap between consistent and inconsistent runs. Anonymous (2026b) extended this to SWE-bench, revealing that consistency amplifies outcomes rather than guaranteeing correctness. Self-consistency (Wang et al., 2023) leverages output-level variance via majority voting but does not examine internal representations.

Probing LLM representations. Hidden states encode truth (Burns et al., 2023; Marks & Tegmark, 2024), model confidence (Kadavath et al., 2022), and answer correctness in reasoning models (Stickland et al., 2024). Kadavath et al. (2024) showed that lightweight probes can detect incorrect generations from hidden states in single-turn settings. We extend probing to *multi-step agentic settings* and find it predicts consistency but, notably, not per-run correctness.

Representation engineering. Zou et al. (2023) introduced representation reading and control; Turner et al. (2023) demonstrated activation steering for behavioral modification. Templeton et al. (2024) extracted "persona vectors" encoding character traits, and Anthropic

(2024) identified an "assistant axis" governing default model behavior. Our work identifies a new representational property—commitment—that may constitute another such axis in activation space.

3 Method

Agent and task. We deploy Llama-3.1-70B-Instruct (Grattafiori et al., 2024) in a ReAct (Yao et al., 2022) agent on HotpotQA (Yang et al., 2018). The agent iterates Thought → Action → Observation using Search, Retrieve, and Finish tools. We select 100 questions from the validation set: 50 "easy" (comparison questions with yes/no answers) and 50 "hard" (multi-hop requiring 2+ retrieval steps). Each question is run 10 times with temperature T=0.5 and maximum 25 steps, yielding 988 total trajectories.[1] At step 4 (the primary analysis step), 99 of 100 questions have sufficient data; one question's runs all terminated by step 3.

Hidden state extraction. The model runs on 8×80GB GPUs with pipeline parallelism (10 layers per GPU). At every agent step, we extract the hidden state at the last token position from layers $\ell \in \{0, 8, 16, \ldots, 80\}$, producing $\mathbf{h}_\ell^{(s)} \in \mathbb{R}^{8192}$ per step s and layer ℓ.

Behavioral consistency. We measure the coefficient of variation (CV) of step counts across the 10 runs of each question. Lower CV indicates more consistent behavior.

Activation similarity. For each question q at step s and layer ℓ, we compute the mean pairwise cosine similarity across runs:

$$\mathrm{Sim}_q^{(s,\ell)} = \binom{n}{2}^{-1} \sum_{i<j} \cos\left(\mathbf{h}_{\ell,i}^{(s)},\ \mathbf{h}_{\ell,j}^{(s)}\right) \tag{1}$$

where n is the number of runs with hidden states at step s. We examine the Pearson correlation between $\mathrm{Sim}_q^{(s,\ell)}$ and CV_q across all questions.

4 Results

4.1 Activation Similarity Predicts Consistency

Figure 1 shows the correlation between activation similarity and behavioral CV across all step–layer combinations. A negative correlation emerges at step 4 across layers 32–80, peaking at layer 40 ($r = -0.348$, 95% CI $[-0.48, -0.17]$, $p = 0.0006$). A 10,000-iteration permutation test confirms significance at all seven sampled layers from 32 to 80 (all permutation $p < 0.002$; full results in Table 2, Appendix B).

Why does high similarity predict inconsistency? By step 4, each run has received different observations (search results, retrieved documents). When hidden states remain similar despite these different inputs, the model has not incorporated evidence into its internal state—it remains uncommitted, and trajectories scatter. When states differentiate, each run has committed to a specific interpretation and proceeds consistently.

4.2 Temporal Profile

Figure 2 shows the correlation at layer 40 across steps. The signal is absent at steps 1–2, strengthens at step 3, peaks at step 4 ($r = -0.348$), and weakens at step 5. This trajectory aligns with Anonymous (2026a), who found that 69% of behavioral divergence begins at step 2. The representational commitment signal appears two steps later, capturing the model's response to the accumulated evidence.

[1] 12 runs excluded due to incomplete hidden state extraction.

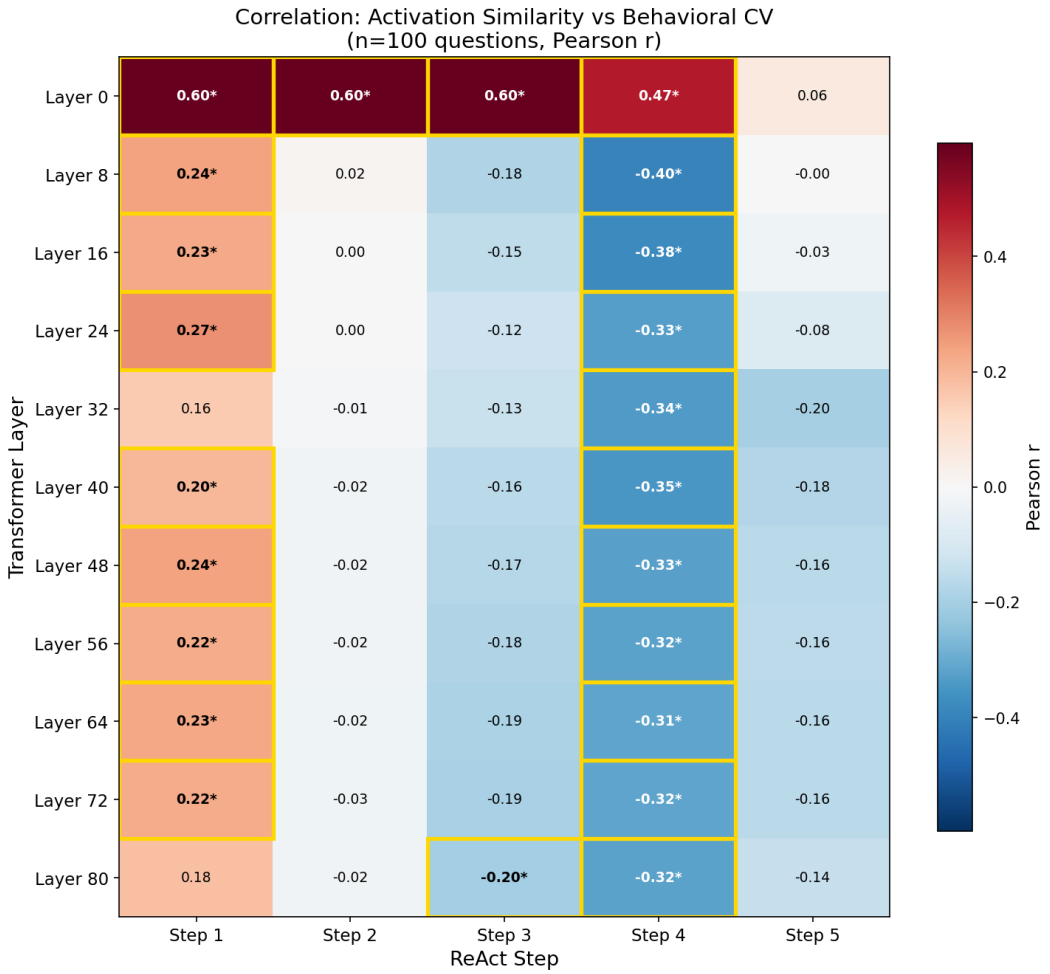

Figure 1: Pearson r between activation similarity and behavioral CV across steps and layers ($n = 99$; one question excluded at step 4 as all runs terminated before this step). Gold borders indicate $p < 0.05$. The signal concentrates at step 4 across layers 32–80.

4.3 Controlling for Task Difficulty

Both activation similarity and CV could be driven by task difficulty. Partial correlation controlling for accuracy and difficulty label *increases* the signal from $r = -0.35$ to $r = -0.45$ (95% CI $[-0.59, -0.29]$, $p < 10^{-5}$) at layer 40, with all layers 32–80 strengthening (partial r: -0.38 to -0.46; Appendix C). Difficulty was partially *suppressing* the signal, confirming it reflects a genuine representational property independent of task characteristics.

4.4 Ruling Out Simple Baselines

We compare hidden state similarity against simpler potential predictors of consistency (full results in Table 5, Appendix F). No input-level feature—question length ($r = 0.10$, $p = .31$), context size ($r = 0.04$, $p = .66$), or thought length ($r = 0.12$, $p = .24$)—predicts CV. A multiple regression (CV $\sim$ similarity + question length + accuracy) yields $R^2 = 0.30$; similarity ($t = -4.73$, $p < 10^{-5}$) and accuracy ($t = -4.41$, $p < 10^{-4}$) are the only significant predictors.

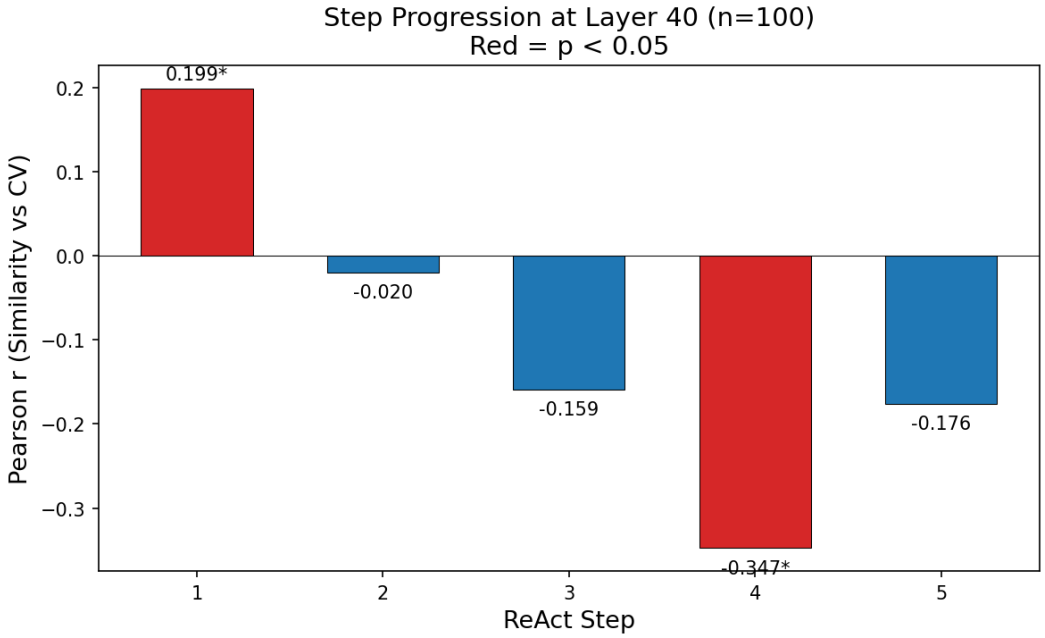

Figure 2: Correlation between activation similarity and CV at layer 40 across steps. The consistency signal peaks at step 4.

4.5 Boundaries of the Signal

Per-run correctness. Linear probes fail to predict per-run correctness from hidden states (AUC 0.34–0.56 across all configurations including single-turn generation), because correctness is primarily question-determined (64 questions are 10/10 correct, 23 are 0/10). The consistency signal is a *between-run relational* property, not a within-run absolute one.

Hard vs. easy questions. The signal is driven by easy questions ($r = -0.57$, $p < 10^{-5}$) rather than hard ($r = -0.02$, $p = 0.88$). Among hard questions, we identify three commitment categories: *committed-correct* (accuracy ≥ 0.8), *committed-wrong* (accuracy ≤ 0.2, CV ≤ 0.15), and *uncommitted-wrong* (accuracy ≤ 0.2, CV > 0.15). Committed-wrong questions show activation similarity indistinguishable from committed-correct (Llama: 0.935 vs. 0.903, $p = 0.30$; Qwen: 0.968 vs. 0.952, $p = 0.46$). Representational commitment is *correctness-agnostic*: the model converges on stable representations whether right or wrong, collapsing the linear relationship within hard questions (Figure 3; Appendix E).

Answer agreement. Activation similarity does not predict answer agreement ($r = -0.13$, $p = 0.22$), confirming the signal reflects *process* consistency (trajectory structure) rather than *output* agreement.

4.6 Cross-Model Validation

We replicate on Qwen 2.5 72B (Yang et al., 2024) (same depth/dimension as Llama, different architecture and training) and Phi-3-Medium-14B (Abdin et al., 2024) (40 layers, d=5120), running all 100 questions $\times$ 10 trials per model (988 Llama, 998 Qwen, 1,000 Phi-3 trajectories).

The negative correlation replicates in both. Qwen shows a stronger peak ($r = -0.65$ at layer 64, 95% CI $[-0.74, -0.55]$, $p < 10^{-6}$). Phi-3 replicates at step 4 ($r = -0.36$ at layer 16, 95% CI $[-0.52, -0.17]$, $p = 0.0005$, n=91) with a stronger signal at step 5 ($r = -0.58$, $p < 10^{-6}$)—the delay plausibly reflecting the smaller model needing additional evidence

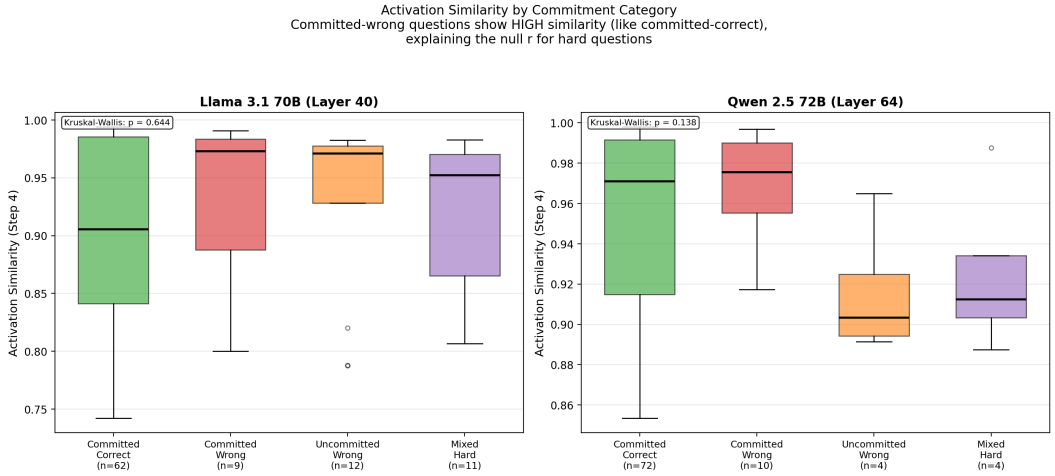

Figure 3: Activation similarity at step 4 by commitment category. Committed-wrong questions (red) show similarity comparable to committed-correct (green), while uncommitted-wrong (orange) trend lower.

before committing. All sampled layers 4–36 are significant at step 4 (all $p < 0.04$; full results in Table 4, Appendix D).[2]

The *peak layer* differs across models—Llama: 50% depth, Qwen: 80%, Phi-3: 40%—suggesting representational commitment is universal but its depth profile is architecture-dependent (Figure 4; temporal profiles in Figure 6, Appendix D).

4.7 Cross-Benchmark Generalization: StrategyQA

We replicate on StrategyQA (Geva et al., 2021) (implicit multi-step reasoning, yes/no answers; 50 questions balanced by answer, $\geq$2 decomposition steps, 10 runs each, T=0.5). The signal replicates strongly at step 3: $r = -0.83$ (95% CI $[-0.90, -0.76]$, $p < 10^{-13}$), spanning all layers 8–80 ($|r| > 0.72$, $p < 10^{-8}$). The one-step-earlier peak is consistent with shorter reasoning chains (mean 5.1 vs. $\sim$12 steps). Partial correlation controlling for accuracy is unchanged ($r = -0.83$), and by step 4 the signal has dissipated ($r \approx 0.07$, $p > 0.7$)—commitment has already occurred (full results in Appendix H).

4.8 Runtime Monitor: Predicting Consistency from Hidden States

We evaluate whether the commitment signal is strong enough for a practical runtime monitor. We frame consistency prediction as binary classification using LOOCV with logistic regression, testing five hidden-state features and four surface-level baselines under two labeling schemes: *quintile* (top/bottom 20% by CV) and *median split* (full results in Table 6, Appendix G).

Hidden-state features achieve strong discrimination. Under quintile labeling, a per-layer similarity profile reaches AUROC = 0.97 [0.90, 1.00] on Llama, and PCA-reduced hidden states reach 0.93 [0.85, 0.99] on Qwen. Under median split, PCA features achieve 0.85 on Llama and layer-averaged similarity reaches 0.88 on Qwen. All surface-level baselines remain near chance (AUROC $\leq$ 0.65), confirming the signal resides in hidden states. Figure 5 shows ROC curves.

[2]The positive correlation at layer 0 ($r = +0.47$ for Llama, $+0.49$ for Qwen; Table 4) reflects input-level similarity: at the embedding layer, hidden states are determined by the (identical) input tokens, so higher similarity indexes shorter, simpler questions that also tend to be more consistent. Phi-3 lacks this pattern ($r = -0.05$, $p = .63$), likely due to its different tokenizer.

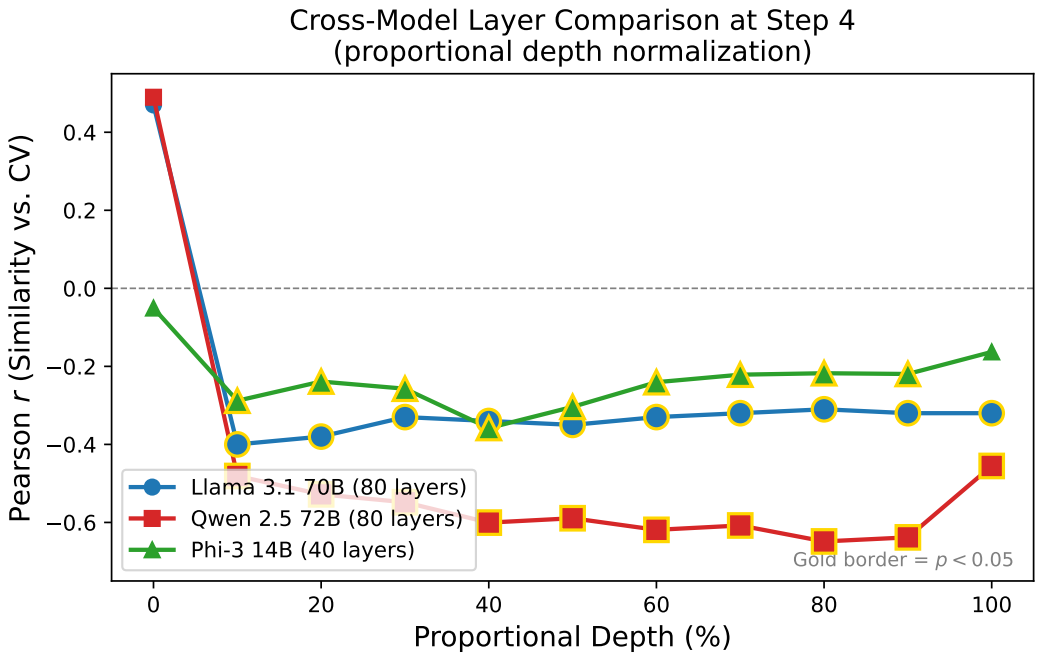

Figure 4: Layer-wise correlation between activation similarity and behavioral CV at step 4 for all three models, with layers normalized to proportional depth (layer / max layer $\times$ 100%) to enable comparison across architectures. All three models show the negative correlation characteristic of representational commitment, with Qwen strongest ($r = -0.65$) and Phi-3 weakest but significant ($r = -0.36$). The positive correlation at 0% depth (embedding layer) for Llama and Qwen reflects an input-similarity artifact absent in Phi-3 (see text). Gold borders: $p < 0.05$.

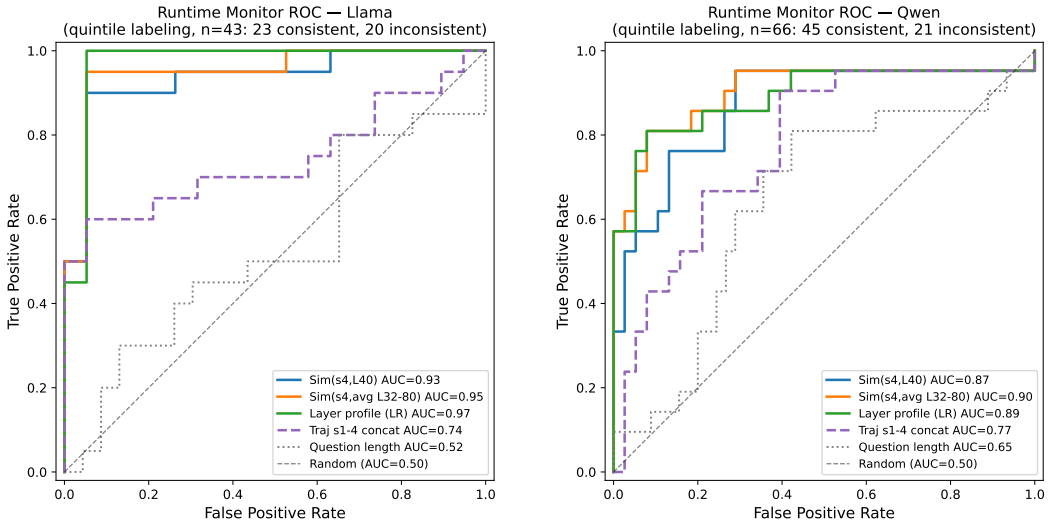

Figure 5: ROC curves for runtime consistency prediction (quintile labeling). Hidden-state features (solid) substantially outperform surface baselines (dashed).

4.9 Intervention: Inducing Representational Commitment

If representational commitment causally drives behavioral consistency, then inducing commitment should reduce downstream variance. We test this with a prompting intervention: at step 3 of the ReAct trajectory—one step before the commitment juncture—we append an instruction to the agent's observation requiring it to explicitly commit to a reasoning strategy before proceeding.[3] The control condition uses the standard ReAct agent with no modification.

Because the commitment prompt adds tokens to the context window, we include a *filler control*: a semantically neutral prompt of matched token length ("Please continue with the task as you normally would...") appended at the same step.[4] This isolates the effect of commitment framing from mere token-count effects.

We run all three conditions on 100 HotpotQA questions (the same set used in the Qwen cross-model validation) with 10 runs per question per condition on Llama 3.1 70B ($T = 0.5$). Table 1 reports the results.

Table 1: Three-condition prompting intervention ($n = 100$ questions, 10 runs each). The filler control matches the commitment prompt in token count but uses semantically neutral text. The critical comparison (Filler vs. Commitment) isolates the commitment framing effect from token-count confounds.

Metric	**Control**	**Filler**	**Commitment**
Behavioral CV	0.112	0.132	0.095
Action seq. diversity	0.250	0.295	0.223
Accuracy	0.836	0.815	0.824
Pairwise comparisons (paired t-test):			
	Filler vs. Commitment (key test)		
CV reduction	28% ($d = 0.33$, $p = .001$)		
Diversity reduction	24% ($d = 0.47$, $p < .001$)		
Accuracy difference	n.s. ($d = 0.11$, $p = .261$)		
	Control vs. Filler		
CV change	+18% ($d = 0.18$, $p = .071$, n.s.)		
	Control vs. Commitment		
CV reduction	15% ($d = 0.20$, $p = .044$)		

The filler control does not reduce CV relative to control ($p = .071$), confirming that additional tokens alone do not improve consistency. The commitment prompt reduces CV by 28% relative to filler (paired $t(99) = 3.29$, $p = .001$, $d = 0.33$; Wilcoxon $p < .001$), establishing that commitment *framing*—not token count—drives the effect. Action diversity shows the same pattern (24% reduction, $d = 0.47$, $p < .001$). No condition affects accuracy (all $p > .05$).

Stratified analysis. The effect concentrates on originally-inconsistent questions (top CV tertile, $n = 32$): filler-vs-commitment ΔCV $= 0.069$ ($d = 0.44$, $p = .018$), while already-consistent questions show a smaller, non-significant trend ($d = 0.27$, $p = .102$)—precisely what a commitment mechanism predicts.

A preliminary activation steering experiment (Appendix A) yielded mixed results: single-layer perturbation at step 4 is too late and too local, suggesting that representational com-

[3]The full prompt reads: "Based on the evidence you have gathered so far, commit to a specific reasoning strategy for solving this question. State your committed strategy clearly in your next Thought, then follow through with it. Do not change strategies or start over—build on what you have learned."

[4]The full filler prompt reads: "Please continue with the task as you normally would. Take the time you need to work through the problem. Consider the information you have gathered and proceed with your next step in whatever way seems most appropriate to you. There is no particular urgency—work at your own pace and follow your reasoning."

mitment is a distributed process unfolding across layers and steps rather than a localized activation pattern.

5 Discussion

Representational commitment as a new axis. Recent work has identified linear directions encoding persona (Templeton et al., 2024), default behavior (Anthropic, 2024), and truth (Marks & Tegmark, 2024). Commitment—the degree to which a model integrates observations—may constitute another such axis, but unlike these static properties, it emerges *dynamically* during multi-step interaction. The prompting intervention (Section 4.9) provides causal evidence.

Why step 4? By step 4, the model has received 3–4 observations and must integrate them. Models that update their representations proceed consistently; those that do not, diverge. The signal fading at step 5 suggests trajectories have already committed or scattered.

Single-turn vs. multi-step probing. Correctness probes that work in single-turn settings (Kadavath et al., 2024; Stickland et al., 2024) fail here (AUC = 0.35 even for single-turn on the same questions), because correctness is question-determined rather than run-determined—a gap in the probing literature that assumes within-input variance exists.

Cross-model and cross-benchmark universality. The different peak layers across models (Llama: 50% depth; Qwen: 80%; Phi-3: 40%) suggest *where* commitment manifests is architecture-dependent, but *that* it manifests is universal. StrategyQA validation ($r = -0.83$) confirms the commitment juncture adapts to task structure.

Limitations. We validate on three models (14B–72B) across two benchmarks; broader task coverage (code generation, mathematical reasoning) would strengthen the claims. CV of step counts is a coarse proxy; sequence-level metrics may reveal finer structure. Although the three-condition intervention rules out token-count confounds, activation-level steering would provide cleaner causal evidence (Appendix A). All models are instruction-tuned; base model behavior is unknown.

Future directions. Three directions follow: (1) circuit tracing at divergence points using attribution graphs (Ameisen et al., 2025) to identify the computations underlying representational commitment; (2) multi-layer activation steering to achieve consistency improvements without prompt modifications; and (3) consistency-aware training via DPO on committed vs. uncommitted trajectories.

6 Conclusion

We introduced *representational commitment*—the degree to which hidden states differentiate across runs—and showed it predicts behavioral consistency in LLM agents ($r = -0.35$, partial $r = -0.45$ after difficulty controls, $4.1\times$ quartile effect). The signal predicts consistency but not correctness, is not explained by simple baselines, and generalizes across three architectures (Llama 70B, Qwen 72B, Phi-3 14B) and two benchmarks (HotpotQA, StrategyQA). A runtime monitor achieves AUROC up to 0.97, and a three-condition intervention confirms that commitment framing—not token count—reduces variance by 28% ($d = 0.33$, $p = .001$). These findings demonstrate that representational commitment is both a measurable phenomenon and a practically exploitable signal for improving agent reliability.

Author Note. Large language models (specifically Claude) were used for code generation, data analysis scripting, and writing assistance during the preparation of this manuscript. All scientific claims, experimental designs, and interpretations were made by the authors.

References

Marah Abdin, Sam Ade Jacobs, Ammar Ahmad Amin, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.

Emmanuel Ameisen, Tom Conerly, et al. Circuit tracing: Revealing computational graphs in language models. *Anthropic Research Blog*, 2025.

Anonymous. When agents disagree with themselves: Measuring behavioral consistency in LLM-based agents. *arXiv preprint arXiv:2602.11619*, 2026a.

Anonymous. Consistency amplifies: How behavioral variance shapes agent accuracy. *Manuscript in preparation*, 2026b.

Anthropic. The Assistant axis: Understanding default model behavior through linear representations. *Anthropic Research Blog*, 2024.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2023.

Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. Did Aristotle use a laptop? A question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361, 2021.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aieleen Letman, Akhil Mathur, Alan Schelten, Amy Yang, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Saurav Kadavath et al. Knowing when you don't know: Detecting incorrect generations from language models using internal representations. *arXiv preprint*, 2024.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2024.

Asa Cooper Stickland, Ethan Perez, et al. Reasoning models know when they're right. *Anthropic Research Blog*, 2024.

Adly Templeton et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Anthropic Research Blog*, 2024.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Leela Castricato. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 2022.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Wang, Bowen Zheng, Chengyuan Yu, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W Cohen, Ruslan Salakhutdinov, and Christopher D Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. *arXiv preprint arXiv:1809.09600*, 2018.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Preliminary Steering Experiment

If representational commitment has a directional signature in activation space, it should be possible to steer the model toward more committed states directly. For 5 questions, we extract the "commitment direction" as the difference between mean hidden states of committed (consistent-correct) and uncommitted (inconsistent) runs at step 4, layer 40, then add this direction (scaled by $\alpha = 1.5$) to hidden states during inference.

Results are mixed: on 2/5 questions, steering reduced step-count CV (from 0.31 to 0.18 and 0.27 to 0.12), while on 3/5 questions it had no effect or slightly increased variance, occasionally disrupting coherent reasoning. We interpret this as a fundamental asymmetry: the prompting intervention succeeds because it changes what the model *generates* at step 3, shaping hidden states across *all* layers in subsequent forward passes. Steering at a single layer at step 4 intervenes too late and too locally—after the commitment juncture has passed. This suggests representational commitment is a distributed computational process, and multi-layer, multi-step steering (Zou et al., 2023) or representation finetuning may be required.

B Permutation Test Results

Table 2: Permutation test at step 4 (10,000 iterations). All layers 32–80 are significant at $p < 0.002$.

Layer	r	Param. p	Perm. p
32	−0.340	0.0008	0.0007
40	**−0.348**	**0.0006**	**0.0003**
48	−0.326	0.0013	0.0012
56	−0.320	0.0017	0.0011
64	−0.313	0.0021	0.0014
72	−0.316	0.0019	0.0019
80	−0.319	0.0017	0.0019

C Partial Correlation Results

Table 3: Partial correlations at step 4, controlling for accuracy and difficulty label. The signal strengthens after removing difficulty-related variance.

Layer	Raw r	Raw p	Partial r	Partial p
32	−0.340	.0008	−0.456	$< .0001$
40	**−0.348**	**.0006**	**−0.445**	$<$ **.0001**
48	−0.326	.0013	−0.416	$< .0001$
56	−0.320	.0017	−0.399	$< .0001$
64	−0.313	.0021	−0.387	.0001
72	−0.316	.0019	−0.387	.0001
80	−0.319	.0017	−0.375	.0002

D Full Cross-Model Layer-Wise Results

Table 4 reports the full layer-wise correlations between activation similarity and behavioral CV at step 4 for all three models. Phi-3 layers are mapped to proportionally equivalent depths (e.g., Phi-3 layer 16 $\approx$ 40% depth).

Figure 6 shows the temporal profile of the commitment signal across all three models. Each model is plotted at its peak layer. The 70B models (Llama and Qwen) reach their strongest signal at step 4, while Phi-3 (14B) peaks one step later at step 5, consistent with the smaller model requiring additional evidence before committing.

Table 4: Cross-model validation at step 4. All three models show the negative correlation between activation similarity and behavioral CV. Phi-3 (14B, 40 layers, d=5120) validates on a structurally different architecture. $^*p < 0.05$.

	Llama 3.1 70B		Qwen 2.5 72B		Phi-3 14B	
Layer	r	p	r	p	r	p
0	0.47*	< .001	0.49*	< .001	−0.05	.634
8	−0.40*	< .001	−0.48*	< .001	−0.24*	.023
16	−0.38*	< .001	−0.53*	< .001	**−0.36***	**< .001**
24	−0.33*	.001	−0.55*	< .001	−0.24*	.022
32	−0.34*	< .001	−0.60*	< .001	−0.22*	.038
40	**−0.35***	**< .001**	−0.59*	< .001	−0.16	.123
48	−0.33*	.001	−0.62*	< .001	—	—
56	−0.32*	.002	−0.61*	< .001	—	—
64	−0.31*	.002	**−0.65***	**< .001**	—	—
72	−0.32*	.002	−0.64*	< .001	—	—
80	−0.32*	.002	−0.46*	< .001	—	—
Sig. layers	11 / 11		11 / 11		4 / 6	
Peak r	−0.35 (L40)		−0.65 (L64)		−0.36 (L16)	
Accuracy	0.70		0.81		0.69	

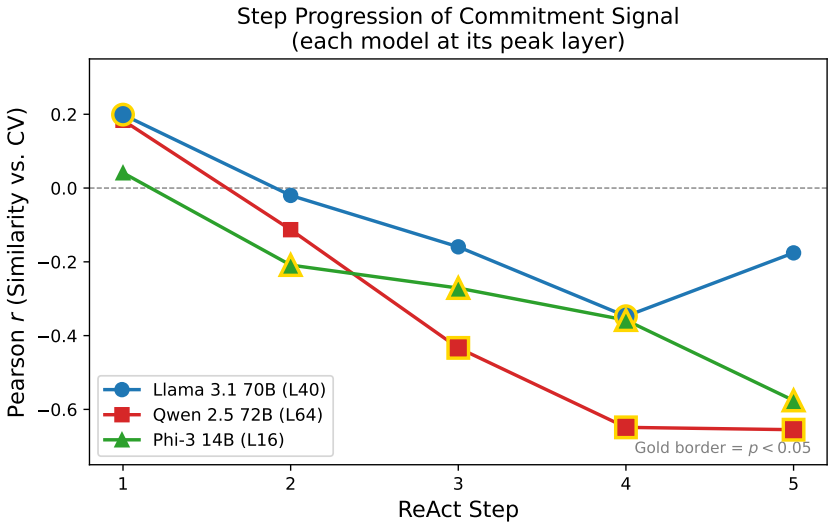

Figure 6: Step progression of the commitment signal across three models (each at its peak layer: Llama L40, Qwen L64, Phi-3 L16). Both 70B models peak at step 4 while Phi-3 (14B) peaks at step 5, suggesting smaller models require one additional evidence-gathering step before committing. Gold borders: $p < 0.05$.

E Commitment Category Visualization

Figure 7 shows a t-SNE visualization of step-4 hidden states (layer 40) for all 100 questions, colored by commitment category. Left panel: Llama 3.1 70B; right panel: Qwen 2.5 72B. Hidden states are first reduced to 50 dimensions via PCA, then projected to 2D via t-SNE (perplexity 30). Each point represents one question's mean hidden state across its 10 runs. Committed-correct (green) and committed-wrong (red) questions occupy overlapping regions of the representation space, consistent with our finding that commitment is correctness-agnostic (Section 4.4). Uncommitted-wrong questions (orange) tend toward more diffuse regions, reflecting their higher activation variance.

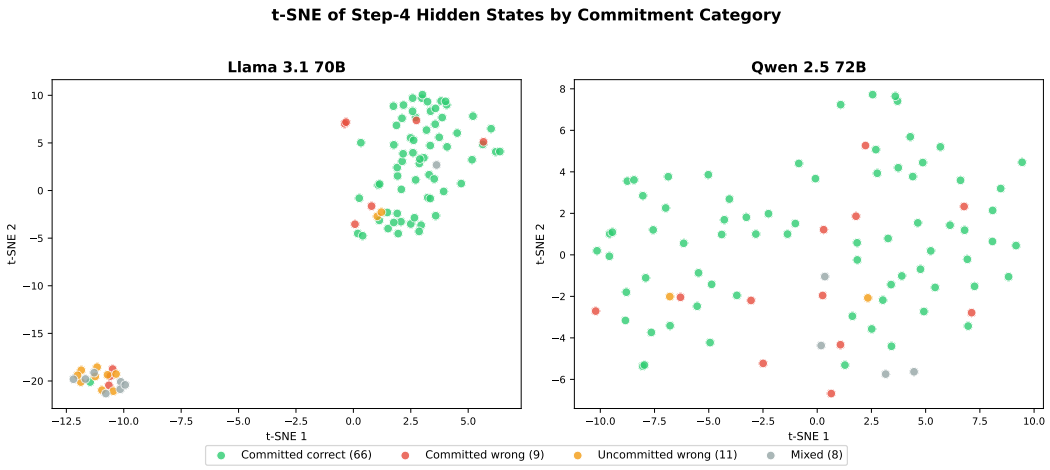

Figure 7: t-SNE projection of step-4 hidden states (PCA to 50D, then t-SNE to 2D) colored by commitment category. Committed-correct (green) and committed-wrong (red) overlap, confirming that representational commitment is orthogonal to correctness. Uncommitted-wrong (orange) questions show more dispersed representations.

F Baseline Predictors of Behavioral CV

Table 5 reports the full set of candidate predictors of behavioral CV, including input-level features and downstream behavioral consequences.

Table 5: Predictors of behavioral CV. †Downstream behavioral consequences, not confounds.

Predictor	r	p
Hidden-state sim. (S4, L40)	**−0.35**	**.0006**
Accuracy (correct rate)†	−0.31	.002
Mean total thought length†	0.26	.011
Mean search actions†	0.23	.019
Mean step count†	0.21	.037
Question word count	0.10	.31
Question char count	0.12	.25
N context documents	0.04	.66
Mean thought length (S3)	0.12	.24

G Runtime Monitor: Full Results

Table 6 reports the complete runtime monitor evaluation across all features, models, and labeling schemes.

H StrategyQA Cross-Benchmark Results

Table 7 reports the full layer-wise correlations between activation similarity and behavioral CV at step 3 (the peak step) for Llama-3.1-70B on StrategyQA (n=50). All layers 8–80 are significant at $p < 10^{-8}$, with partial correlations controlling for accuracy virtually unchanged due to the high overall accuracy (93.2%).

Table 8 summarizes cross-benchmark comparison. The StrategyQA signal is stronger than HotpotQA despite a smaller sample, and peaks one step earlier consistent with its shorter reasoning chains.

Table 6: Runtime monitor AUROC (95% CI) for predicting behavioral consistency. Features A–E use step-4 hidden states; baselines use only surface-level input properties.

	Llama 3.1 70B		Qwen 2.5 72B	
Feature	Quintile	Median	Quintile	Median
A: Sim (L40)	.93	.55	.87	.86
B: Sim (avg)	.95	.58	.90	.88
C: Layer profile	**.97**	.73	.89	.88
D: Trajectory	.74	.55	.77	.79
E: PCA (L40)	.94	**.85**	**.93**	.67
Random	.43	.57	.48	.48
Question length	.52	.60	.65	.57
Context docs	.08	.06	.07	.06
Thought length	.10	.04	.00	.00

Table 7: StrategyQA: Layer-wise correlations at step 3 (Llama-3.1-70B, n=50). The signal spans all non-embedding layers and peaks at layer 72. Partial correlations control for accuracy.

Layer	Raw r	Raw p	Partial r	Partial p
0	—	—	—	—
8	-0.73	2.5×10^{-9}	-0.72	2.9×10^{-9}
16	-0.78	3.8×10^{-11}	-0.77	4.3×10^{-11}
24	-0.76	2.3×10^{-10}	-0.75	3.7×10^{-10}
32	-0.78	2.6×10^{-11}	-0.78	4.0×10^{-11}
40	-0.81	8.7×10^{-13}	-0.81	8.4×10^{-13}
48	-0.82	2.0×10^{-13}	-0.82	1.9×10^{-13}
56	-0.83	8.8×10^{-14}	-0.83	9.7×10^{-14}
64	-0.83	7.7×10^{-14}	-0.83	9.3×10^{-14}
72	$\mathbf{-0.83}$	$\mathbf{7.4 \times 10^{-14}}$	$\mathbf{-0.83}$	$\mathbf{9.1 \times 10^{-14}}$
80	-0.82	4.8×10^{-13}	-0.82	4.7×10^{-13}

Figure 8 shows the step$\times$layer heatmap of Pearson correlations between activation similarity and behavioral CV for StrategyQA, revealing the sharp commitment signal concentrated at step 3 across all layers.

Table 8: Cross-benchmark comparison of the representational commitment signal (Llama-3.1-70B). StrategyQA produces a stronger signal that peaks one step earlier, consistent with shorter reasoning chains.

	HotpotQA	StrategyQA
Task type	Multi-hop retrieval QA	Implicit reasoning
Answer type	Span extraction	Yes/no
n (questions)	100	50
Runs per question	10	10
Mean chain length	∼12 steps	5.1 steps
Accuracy	70%	93.2%
Peak step	4	3
Peak layer	40	72
Peak r	-0.35	-0.83
95% CI	$[-0.48, -0.17]$	$[-0.90, -0.76]$
Peak p	< 0.001	$< 10^{-13}$
Partial r	-0.45	-0.83
Sig. layers	7 / 11	10 / 11

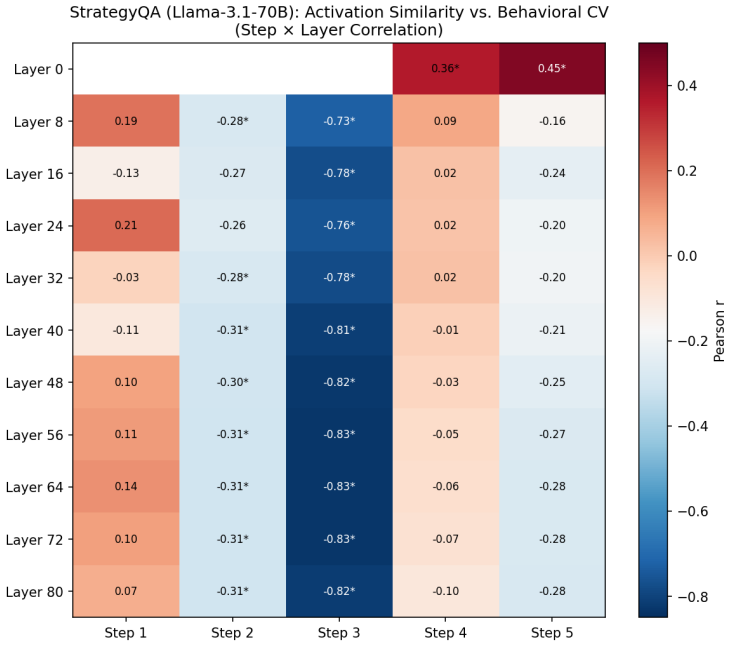

Figure 8: StrategyQA step×layer correlation heatmap (Llama-3.1-70B, n=50). The commitment signal concentrates at step 3 (deep blue column), one step earlier than HotpotQA's step 4 peak, consistent with shorter reasoning chains. All layers 8–80 at step 3 show $|r| > 0.72$ ($p < 10^{-8}$). $^*p < 0.05$.